

Experiment 4: Implementation of Fuzzy Set Operations

Objective

Write a Program to implement the Union, Intersection, Complement, and Difference operations on fuzzy sets.

Theory

A fuzzy set is characterized by a membership function that assigns each element of the universe of discourse a value between 0 and 1.

Let A and B be two fuzzy sets defined on the same universe X .

- **Union of fuzzy sets**

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

- **Intersection of fuzzy sets**

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

- **Complement of a fuzzy set**

$$\mu_{\neg A}(x) = 1 - \mu_A(x)$$

- **Difference of fuzzy sets**

$$\mu_{A-B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$$

Code

```
%matplotlib inline
import matplotlib.pyplot as plt

def fuzzy_union(A, B):
    return {x: max(A[x], B[x]) for x in A}

def fuzzy_intersection(A, B):
    return {x: min(A[x], B[x]) for x in A}

def fuzzy_complement(A):
    return {x: 1 - A[x] for x in A}

def fuzzy_difference(A, B):
    return {x: min(A[x], 1 - B[x]) for x in A}

# Fuzzy sets
A = {'x1': 0.2, 'x2': 0.7, 'x3': 1.0}
B = {'x1': 0.5, 'x2': 0.4, 'x3': 0.6}

U = list(A.keys())

# Plot fuzzy sets A and B
plt.figure()
plt.plot(U, list(A.values()), marker='o', label='A')
plt.plot(U, list(B.values()), marker='s', label='B')
plt.xlabel("Universe of Discourse")
plt.ylabel("Membership Value")
plt.title("Fuzzy Sets A and B")
plt.legend()
plt.grid(True)
plt.show()

# Plot union
plt.figure()
plt.plot(U, list(fuzzy_union(A, B).values()), marker='o')
plt.title("Union of A and B")
plt.xlabel("Universe of Discourse")
plt.ylabel("Membership Value")
plt.grid(True)
plt.show()

# Plot intersection
plt.figure()
plt.plot(U, list(fuzzy_intersection(A, B).values()), marker='o')
plt.title("Intersection of A and B")
plt.xlabel("Universe of Discourse")
plt.ylabel("Membership Value")
plt.grid(True)
plt.show()
```

```
# Plot complement
plt.figure()
plt.plot(U, list(fuzzy_complement(A).values()), marker='o')
plt.title("Complement of A")
plt.xlabel("Universe of Discourse")
plt.ylabel("Membership Value")
plt.grid(True)
plt.show()

# Plot difference
plt.figure()
plt.plot(U, list(fuzzy_difference(A, B).values()), marker='o')
plt.title("Difference of A and B (A - B)")
plt.xlabel("Universe of Discourse")
plt.ylabel("Membership Value")
plt.grid(True)
plt.show()
```

Output

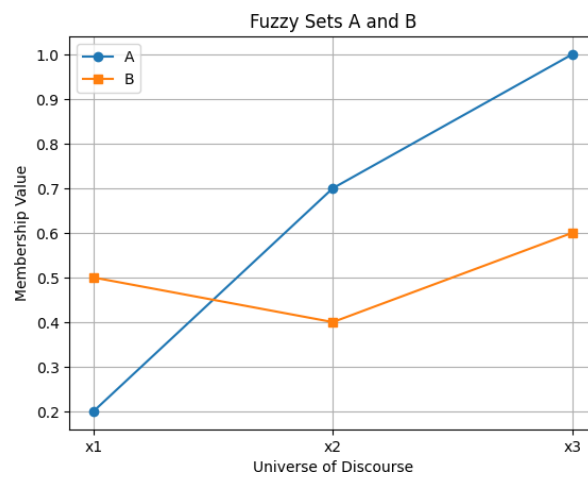


Figure 1: Membership Functions of Fuzzy Sets A and B

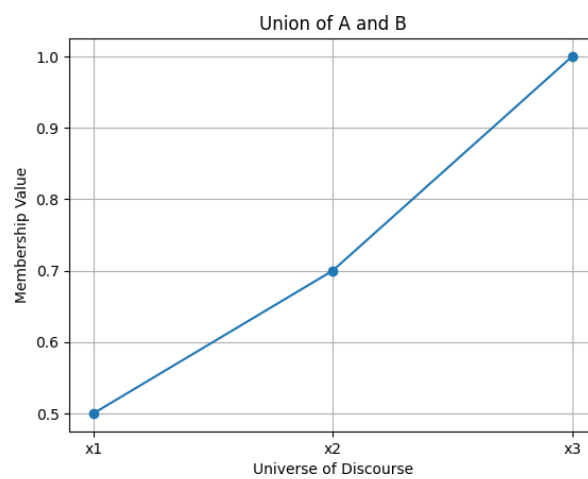


Figure 2: Union of Fuzzy Sets A and B

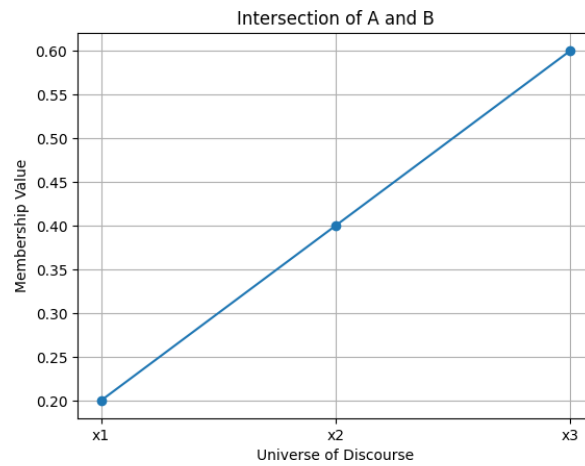


Figure 3: Intersection of Fuzzy Sets A and B

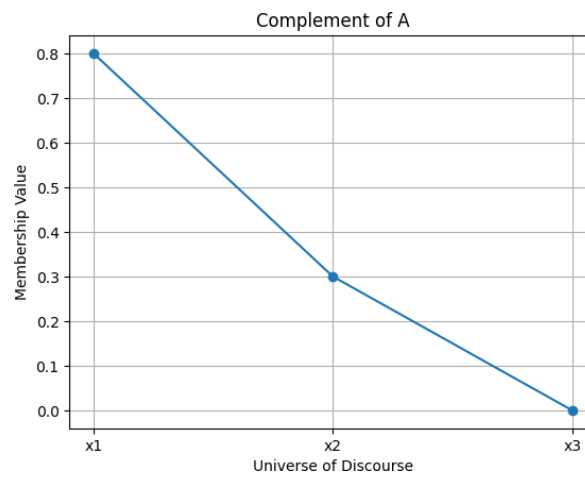


Figure 4: Complement of Fuzzy Set A

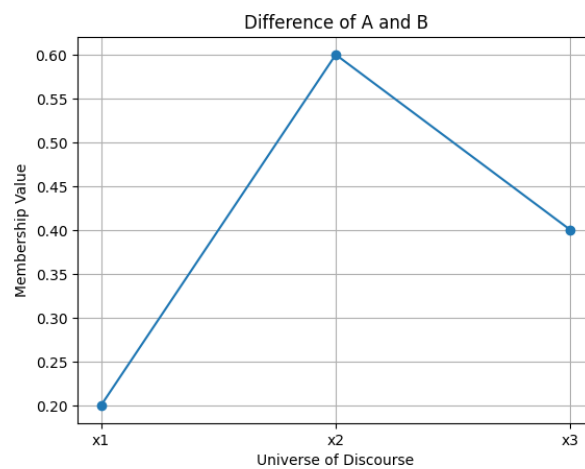


Figure 5: Difference of Fuzzy Sets (A B)

Conclusion

Thus, the fuzzy set operations Union, Intersection, Complement, and Difference were successfully implemented. The results demonstrate the effectiveness of fuzzy logic in handling partial membership values.